

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	Sree B	Reg. No.:	192524023
Section / Batch:	AI and DS / 2025	Date:	24/07/2026

Code of Conduct

I Sree B (192524023) _____ (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sree B _____

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 - 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:			100 Marks

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

SET-1

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)
Back

QUESTIONS

Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack - Balancing Symbols 5 marks

Q6 Singly Linked List 5 marks

Q7 Stack 5 marks

Q8 Linked List 5 marks

Q9 Linked List 5 marks

Q10

10/10 answered

Q1 Binary Search 5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {
    int low = 0, high = n - 1;
    while(low < high) {
        int mid = (low + high) / 2;
        if(arr[mid] == key)
            return mid;
        else if(arr[mid] < key)
            low = mid;
        else
            high = mid;
        }
        return -1;
    }
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int binarySearch(int arr[], int n, int key) {
2     int low = 0, high = n - 1;
3
4     while (low <= high) {
5         int mid = low + (high - low) / 2;
6
7         if (arr[mid] == key)
8             return mid;
9         else if (arr[mid] < key)
10            low = mid + 1;
11        else
12            high = mid - 1;
13    }
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:/C/CP/Comp/CCP/1/bin/.../lib/gcc/
/x86_64-w64-
mingw32/10.3.5/.../lib/x86_64-
4-w64-mingw32/bin/ld.exe:
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)
Back

QUESTIONS

Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack - Balancing Symbols 5 marks

Q6 Singly Linked List 5 marks

Q7 Stack 5 marks

Q8 Linked List 5 marks

Q9 Linked List 5 marks

Q10

10/10 answered

Q2 Stack 5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {
    if(top == 0) {
        printf("Underflow");
        return -1;
    }
    return stack[top--];
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#define MAX 100
int stack[MAX];
int top = -1;

void push(int x) {
    if (top == MAX - 1) {
        printf("Overflow\n");
        return;
    }
    stack[++top] = x;
}

int pop() {
    if (top == -1) {
        printf("Underflow\n");
        return -1;
    }
    return stack[top--];
}

int main() {
    int n, x, choice;

    printf("Enter number of operations: ");
    scanf("%d", &n);

    for (int i = 0; i < n; i++) {
        printf("Enter choice (1:push, 2:pop): ");
        scanf("%d", &choice);

        if (choice == 1) {
            printf("Enter value to push: ");
            scanf("%d", &x);
            push(x);
        } else if (choice == 2) {
            int val = pop();
            if (val != -1) printf("Popped: %d\n", val);
        }
    }
}
```

Test Input

1
10
1
20
2

Output

```
Enter choice to push/pop:
Enter value to push: Enter
choice (1:push, 2:pop): Enter
value to push: Enter choice
(1:push, 2:pop): Popped: 20
```


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3 CREDITS

100% COMPLETION

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack

1 marks

✓

Q2

Circular Queue

1 marks

✓

Q3

Stack

1 marks

✓

Q4

Stack

1 marks

✓

Q5

Doubly Linked List

1 marks

✓

Q6

Queue

1 marks

✓

Q7

Queue

1 marks

✓

Q8

Binary Search

1 marks

✓

Q9

Circular Queue

1 marks

✓

Q10

Stack

1 marks

✓

10/10 answered

Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {
    if (del->prev != NULL)
        del->prev->next = del->next;
    if (del->next != NULL)
        del->next->prev = del->prev;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node* prev;
    struct Node* next;
};

void deleteNode(struct Node** head, struct Node* del) {
    if (*head == NULL || del == NULL)
        return;

    if (*head == del)
        *head = del->next;

    if (del->prev != NULL)
        del->prev->next = del->next;
    if (del->next != NULL)
        del->next->prev = del->prev;

    free(del);
}

void display(struct Node* head) {
    struct Node* temp = head;
    while (temp != NULL) {
        printf("%d ", temp->data);
        temp = temp->next;
    }
    printf("\n");
}
```

Test Input

stdin input (optional)

Output

Original list: 10 20 30
After deletion: 10 30

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3 CREDITS

100% COMPLETION

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack

1 marks

✓

Q2

Circular Queue

1 marks

✓

Q3

Stack

1 marks

✓

Q4

Stack

1 marks

✓

Q5

Doubly Linked List

1 marks

✓

Q6

Queue

1 marks

✓

Q7

Queue

1 marks

✓

Q8

Binary Search

1 marks

✓

Q9

Circular Queue

1 marks

✓

Q10

Stack

1 marks

✓

10/10 answered

Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {
    if (rear == MAX) return;
    queue[rear++] = data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#define MAX 5

int queue[MAX];
int front = -1;
int rear = -1;

void enqueue(int value) {
    // Overflow condition for circular queue
    if ((front == 0 && rear == MAX - 1) || (rear + 1) % MAX == front)
        printf("Queue Overflow\n");
        return;

    if (front == -1) front = 0;
    rear = (rear + 1) % MAX;
    queue[rear] = value;
    printf("%d inserted into queue\n", value);
}

void dequeue() {
    if (front == -1)
        printf("Queue Underflow\n");
        return;
    printf("Deleted: %d\n", queue[front]);
    // If only one element is present
    if (front == rear)
        front = rear = -1;
    return;
    front = (front + 1) % MAX;
}

void display() {
    // ... (code for displaying the queue) ...
}
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Circular Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Prefix Expression
5 marks

10/10 answered

Queue - Deque

5 marks

Find the error in this Deque InsertFront operation.
void insertFront(int data) {
if (front == 0) front = MAX - 1;
else front = front - 1;
deque[front] = data;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#define MAX 5
int deque[MAX];
int front = -1;
int rear = -1;

void insertFront(int data) {
    if ((front == 0 && rear == MAX - 1) || (front == rear + 1)) {
        printf("Deque is Full\n");
        return;
    }
    if (front == -1) {
        front = 0;
        rear = 0;
    }
    else if (front == 0) {
        front = MAX - 1;
    }
    else {
        front = front - 1;
    }
    deque[front] = data;
}

void insertRear(int data) {
    if ((front == 0 && rear == MAX - 1) || (front == rear + 1)) {
        printf("Deque is Full\n");
        return;
    }
    if (rear == -1) {
        front = 0;
        rear = 0;
    }
    else if (rear == MAX - 1) {
        rear = 0;
    }
    deque[rear] = data;
}

int main() {
    int arr[] = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12};
    int n = sizeof(arr) / sizeof(arr[0]);
    int result = binarySearch(arr, 0, n - 1, 5);
    if (result == -1) {
        printf("Element found at index %d\n", result);
    }
    else {
        printf("Element not found\n");
    }
    return 0;
}
```

Test Input

stdin input (optional)

Output

Deque elements: 1 2 3 4 5

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SHEET 8
18/03/2024 10:58 AM

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Circular Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Prefix Expression
5 marks

10/10 answered

Binary Search

5 marks

Identify the bug in this Recursive Binary Search.
int binarySearch(int arr[], int l, int r, int x) {
int mid = l + (r - l) / 2;
if (arr[mid] == x) return mid;
if (arr[mid] > x) return binarySearch(arr, l, mid, x);
return binarySearch(arr, mid + 1, r, x);
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
int binarySearch(int arr[], int l, int r, int x) {
    if (l > r) return -1;

    int mid = l + (r - l) / 2;
    if (arr[mid] == x) return mid;
    if (arr[mid] > x)
        return binarySearch(arr, l, mid - 1, x);
    else
        return binarySearch(arr, mid + 1, r, x);
}

int main() {
    int arr[] = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12};
    int n = sizeof(arr) / sizeof(arr[0]);
    int x = 5;
    int result = binarySearch(arr, 0, n - 1, x);
    if (result == -1) {
        printf("Element found at index %d\n", result);
    }
    else {
        printf("Element not found\n");
    }
    return 0;
}
```

Test Input

stdin input (optional)

Output

Element found at index 4

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